Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Getting to Know Plane Shapes

A plane shape is a flat shape that lies on a single level surface. You can think of it as a shape you could trace around the edge of, because it only has length and width — it has no height or thickness. The outside of the shape is its boundary, and the space inside that boundary is its area.

Plane shapes are grouped mainly by how many straight sides they have, and shapes with straight sides are called polygons. A polygon with three sides is a triangle, one with four sides is a quadrilateral, five sides a pentagon, six sides a hexagon, and so on. A shape with a curved boundary and no straight sides, like a circle, is not a polygon.

Here are the most common plane shapes and the properties that describe them:

Triangle — a polygon with exactly 3 sides and 3 corners (called vertices). The three sides may be equal or unequal, which gives different special triangles.

Square — a quadrilateral with 4 equal sides and 4 right angles (each angle is 90°). It is a special rectangle, because it has all the properties of a rectangle and one more: all its sides are equal.

Rectangle — a quadrilateral with 4 right angles. Its two opposite sides are equal and parallel, but the four sides are not all equal (unlike the square).

Parallelogram — a quadrilateral with two pairs of parallel, equal sides. Its opposite angles are equal, but its angles are not necessarily right angles.

Trapezium — a quadrilateral with only one pair of parallel sides. (Some books call this shape a trapezoid.)

Circle — a round plane shape with a curved boundary. It has no straight sides, no corners and no vertices. Every point on its boundary is the same distance from its centre.

When you compare shapes, you look for their similarities (things they have in common) and their differences (things that set them apart). Here is how to talk about them clearly.

Similarities among quadrilaterals. A square, rectangle, parallelogram and trapezium are all quadrilaterals, so each one has 4 sides, 4 corners and 4 angles. That is their biggest similarity.

Differences between a square and a rectangle. Both have 4 right angles, so they are similar there. But a square has all four sides equal, while a rectangle has only its opposite sides equal. So every square is a rectangle, but not every rectangle is a square.

Differences between a parallelogram and a trapezium. A parallelogram has two pairs of parallel sides; a trapezium has only one pair. That single difference tells them apart.

How a circle differs from all polygons. A circle has a curved boundary and no straight sides and no corners, while every polygon has straight sides and corners. So the circle is the odd one out among the common plane shapes.

Worked example 1 — Naming a shape from its properties

A shape has 4 straight sides, exactly one pair of those sides are parallel, and none of its angles are right angles. What is the shape?

Look at each clue. It has 4 sides, so it is a quadrilateral. Only one pair of sides is parallel, which is the defining property of a trapezium — a parallelogram would have two pairs of parallel sides. The clue about no right angles is consistent, so the shape is a trapezium.

Worked example 2 — Spotting a similarity and a difference

Compare a square and a parallelogram.

Similarity: Both are quadrilaterals, so each has 4 sides and 4 corners.
Difference: A square has 4 right angles and all sides equal; a parallelogram has no right angles (in general) and only its opposite sides are equal. A square is a special kind of parallelogram because its opposite sides are parallel, but a parallelogram is not a square.

Worked example 3 — Telling a polygon from a non-polygon

Which of these is not a polygon: pentagon, triangle, circle, hexagon?

A polygon is made of straight sides and has corners. The pentagon, triangle and hexagon are all polygons. The circle has a curved boundary and no corners, so it is not a polygon.

Practice

1. A shape has 4 straight sides, all four sides are equal, and all four angles are right angles. What is this shape? Name one other plane shape that is similar to it, and one way they are different.
2. Which of these shapes is not a polygon: triangle, parallelogram, circle, hexagon? Give the reason for your choice.
3. A parallelogram and a trapezium are both quadrilaterals. State one similarity between them and one difference between them.
4. State three properties of a rectangle, and then explain why a square can also be described as a rectangle.

Section 2: The Parts of a Circle

A circle is a special plane shape in which every point on its rim is the same distance from one fixed point at its middle. Because of this, a circle has its own set of named parts. Learning these names lets you describe circles precisely and get ready to measure them.

Here are the main parts of a circle and what each one means.

The centre is the fixed point exactly in the middle of the circle. Every other part of the circle is named in relation to it. We often label it with a letter, such as O.

The circumference is the distance all the way around the circle — it is the circle's outer boundary, the curved rim itself.

The radius (plural: radii) is any straight line that joins the centre to a point on the circumference. Every radius of a circle has the same length. If you double the radius, you get the diameter.

The diameter is a straight line that passes through the centre and touches the circumference at both ends. The diameter is twice the length of the radius, and every diameter divides the circle into two equal halves.

A chord is a straight line that joins two points on the circumference. The diameter is the longest possible chord, because it is the only chord that passes through the centre.

An arc is part of the circumference. A small part is called a minor arc, and the larger remaining part is called a major arc.

A sector is the region of the circle "sliced" between two radii, shaped like a slice of cake, with an arc as its curved edge.

A segment is the region between a chord and the arc it cuts off.

A semicircle is exactly half of a circle, bounded by a diameter and the arc it cuts.

A quadrant is a quarter of a circle, bounded by two perpendicular radii and the quarter-arc between them.

A quick way to remember the relationships: the radius reaches from the centre to the rim, the diameter reaches across the circle through the centre (twice the radius), and the circumference is the whole way around.

Worked example 1 — Naming the parts

Imagine a circle with its centre marked O. A line runs from O to the rim and is labelled r. Another line runs right across the circle through O, touching the rim at both ends, and is labelled d.

What are the lines r and d called, and how are their lengths related?

Solution. The line r joins the centre O to the circumference, so it is a radius. The line d passes through the centre and touches the circumference at both ends, so it is a diameter. The diameter is twice the radius, so d = 2r. Equally, the radius is half the diameter: r = d ÷ 2.

Worked example 2 — Spotting the diameter

A chord of a circle is 6 cm long and passes through the centre of the circle. What kind of chord is it, and what is the radius of the circle?

Solution. A chord that passes through the centre is a diameter. Since the diameter is 6 cm, the radius is half of it: 6 ÷ 2 = 3 cm.

Worked example 3 — Relating radius and diameter

The radius of a circle is 7 cm. Find the length of its diameter.

Solution. The diameter is twice the radius: 2 × 7 = 14 cm.

Practice

1. A circle has a radius of 5 cm. What is the length of its diameter?
2. The diameter of a circle is 18 cm. What is the length of its radius?
3. A straight line joins two points on the circumference of a circle but does not pass through the centre. What is this line called, and what is the name of the longest possible line of this kind?
4. Match each part of a circle to its meaning: (a) the distance all the way around the circle, (b) the fixed point at the middle, (c) half of a circle, (d) the region between two radii and the arc they cut off.

Section 3: Finding Perimeter Around Plane Shapes

The perimeter of a plane shape is the total distance all the way around its outside edge. Think of it as the length of fencing you would need to go right around the shape. Because it is a length, perimeter is always measured in units of length such as centimetres (cm), metres (m) or kilometres (km).

You find the perimeter by adding up the lengths of every side of the shape. For some special shapes, there is a quicker way, because some sides are equal.

Let us meet the shapes you will work with.

A square has four equal sides. If one side is called s, then the perimeter is s + s + s + s = 4s. So Perimeter of a square = 4 × side.

A rectangle has two pairs of equal sides: two lengths and two breadths (widths). If the length is L and the breadth is B, then Perimeter = 2 × (L + B), because the opposite sides match.

A regular polygon is any shape whose sides are all equal and whose angles are all equal — for example a regular pentagon (5 sides), a regular hexagon (6 sides) or a regular octagon (8 sides). If a regular polygon has n equal sides and each side is s long, then Perimeter = n × s.

A trapezium has four sides, but only one pair of opposite sides is parallel. There is no shortcut, so you just add up all four sides: Perimeter = a + b + c + d.

A parallelogram has two pairs of equal, opposite sides. If the two unequal side lengths are a and b, then Perimeter = 2 × (a + b), just like a rectangle.

A circle is different. Its perimeter is called the circumference. A circle does not have straight sides, so it needs its own rule. The circumference depends on how far it is across the middle (the diameter, d) or from the middle to the edge (the radius, r, which is half the diameter). Using the number π (about 3.14), Circumference = π × d = 2 × π × r.

Worked example 1 — a square and a rectangle
A square has sides of 6 cm. Its perimeter is 4 × 6 cm = 24 cm. A rectangle is 10 cm long and 4 cm broad. Its perimeter is 2 × (10 + 4) = 2 × 14 = 28 cm.

Worked example 2 — a regular polygon
A regular hexagon has 6 equal sides, each 5 cm. Perimeter = n × s = 6 × 5 = 30 cm.

Worked example 3 — a trapezium and a parallelogram
A trapezium has sides 3 cm, 5 cm, 4 cm and 6 cm. Perimeter = 3 + 5 + 4 + 6 = 18 cm. A parallelogram has sides 7 cm and 3 cm (with the opposite sides matching). Perimeter = 2 × (7 + 3) = 2 × 10 = 20 cm.

Worked example 4 — a circle
A circle has a diameter of 10 cm. Circumference = π × d = 3.14 × 10 = 31.4 cm. If instead we are given a radius of 4 m, then Circumference = 2 × π × r = 2 × 3.14 × 4 = 25.12 m.

Practice

1. A square garden has sides of 9 m. What is its perimeter?
2. A rectangle is 12 cm long and 7 cm broad. Find its perimeter.
3. A regular pentagon has sides of 8 cm each. What is its perimeter?
4. A trapezium has sides of 4 cm, 6 cm, 5 cm and 7 cm. Find its perimeter.
5. A parallelogram has sides of 9 cm and 5 cm. Find its perimeter.
6. A circle has a radius of 7 cm. Use π = 3.14. What is its circumference?

Check your work

Attempt every question before you read this page.

Getting to Know Plane Shapes

1. The shape is a square. A rectangle is similar to it (both are quadrilaterals with 4 right angles), but they differ because a square has all four sides equal while a rectangle has only its opposite sides equal.

   Four sides means a quadrilateral; all four sides equal and four right angles together define a square. For comparison, the rectangle shares the 4 right angles and the quadrilateral property, but not the equal sides.
2. The circle is not a polygon because it has a curved boundary and no straight sides and no corners, while a polygon is made of straight sides and has corners.

   The triangle, parallelogram and hexagon each have only straight sides and at least three corners, so they are polygons. The circle's boundary is curved all the way round, so it does not fit the definition of a polygon.
3. Similarity: both have 4 straight sides and 4 corners (they are both quadrilaterals). Difference: a parallelogram has two pairs of parallel sides, while a trapezium has only one pair of parallel sides.

   Both fall under "quadrilateral," so 4 sides and 4 corners is their shared property. The number of pairs of parallel sides is what separates them: two pairs for the parallelogram against one pair for the trapezium.
4. A rectangle has 4 sides, 4 right angles, and its opposite sides are equal and parallel. A square can be described as a rectangle because it also has 4 right angles and opposite sides parallel, and it simply adds the property that all four sides are equal — so every square meets the requirements of a rectangle.

   List the rectangle's defining features (4 sides, 4 right angles, opposite sides equal and parallel). Since the square shares all of these features and only adds equal sides, it fits the description of a rectangle, making it a special kind of rectangle.

The Parts of a Circle

1. The diameter is twice the radius, so it is 2 × 5 = 10 cm.

   The diameter is always twice the radius. Multiply the radius by 2: 2 × 5 = 10 cm.
2. The radius is half the diameter, so it is 18 ÷ 2 = 9 cm.

   The radius is always half the diameter. Divide the diameter by 2: 18 ÷ 2 = 9 cm.
3. It is called a chord. The longest possible chord is the diameter (the chord that passes through the centre).

   A straight line joining two points on the circumference is a chord. The diameter is the special chord that passes through the centre, and it is the longest chord a circle can have.
4. (a) circumference, (b) centre, (c) semicircle, (d) sector.

   (a) The distance around the whole circle is the circumference. (b) The fixed middle point is the centre. (c) Half of a circle, bounded by a diameter, is a semicircle. (d) The region between two radii and the arc between them, shaped like a cake slice, is a sector.

Finding Perimeter Around Plane Shapes

1. $4 \times 9 = 36$ m

   A square has four equal sides. Perimeter = 4 × side = $4 \times 9 = 36$ m.
2. $2 \times (12 + 7) = 2 \times 19 = 38$ cm

   Perimeter of a rectangle = $2 \times (L + B) = 2 \times (12 + 7) = 2 \times 19 = 38$ cm.
3. $5 \times 8 = 40$ cm

   A regular pentagon has 5 equal sides. Perimeter = $n \times s = 5 \times 8 = 40$ cm.
4. $4 + 6 + 5 + 7 = 22$ cm

   A trapezium has no equal sides, so add all four: $4 + 6 + 5 + 7 = 22$ cm.
5. $2 \times (9 + 5) = 2 \times 14 = 28$ cm

   A parallelogram has two pairs of equal sides. Perimeter = $2 \times (a + b) = 2 \times (9 + 5) = 2 \times 14 = 28$ cm.
6. $2 \times 3.14 \times 7 = 43.96$ cm

   Circumference = $2 \times \pi \times r = 2 \times 3.14 \times 7 = 6.28 \times 7 = 43.96$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-ce983e35	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____